Appendix F. Detailed results of the selected design output

This appendix presents the detailed results of the selected design output analyzed in Section 5.3.3. Tables 16 – 18 describe the selected run analyzed in Figs. 21 and 22 (and, for Case 1, in Fig. 18 of the main text), and Table 19 lists, for one body of each case, the IDs and coordinates of the panels and segments it contains, corresponding to the ID mapping in Figs. 18 and 22.

As shown in Fig. 21, the BIM model represents the design results of the first stage of the EA, i.e., the PWP layout design. All walls are accurately segmented, and the local zooms show that the segmented PWP walls maintain consistent heights before and after segmentation, not only in simple single-wall scenarios but also in complex multi-corner areas, which illustrates the effectiveness of the first stage of the EA.

Tables 16 and 17 present the statistical results for the optimized design of PWP segments and standard PWP panels, respectively. A comparison of Tables 16 and 10 reveals that the total width of the generated PWP segments matches the cumulative length of the PWP wall and that the areas of both are identical, validating the accuracy of the proposed method in segmenting the PWP wall. In addition, the width of each PWP segment is confined within the predetermined range, confirming that the size constraints of individual segments are met. Comparing Tables 16 and 17, the total width and area of the standard PWP panels exceed those of the corresponding PWP segments, whereas the number of standard panels is smaller than the number of segments; this discrepancy is due to the assembly of segments into standard panels and the material wastage that occurs when filling the segments to form standard panels.

Fig. 22 illustrates the BIM models of Case 2, which, like those of Case 1 in Fig. 18, include the layout model and the manufacturing combination of the second-stage EA, presented from the three perspectives of segments, (standard) panels, and bodies with the hierarchical ID mapping described in Section 5.3.3.

Table 18 summarizes the PWP bodies of the selected run: Case 1 requires eight bodies and Case 2 six, consistent with Figs. 18 and 22. In terms of nominal material consumption, which accounts for all standard panels produced in this cycle, the waste is 4.52% for Case 1 and 9.17% for Case 2. The higher value of Case 2 arises because 72 standard panels are produced but only 69 are needed, leaving three surplus panels (the remaining standard panels in Table 18) that can be used in other projects; after excluding these surplus panels (usable area), the waste of Case 2 drops to 5.14%, whereas Case 1 produces no surplus panels and remains at 4.52%, so the material waste

of both cases is below about 5%, indicating high material efficiency. The higher waste of Case 2 is attributable to its smaller number of PWP segments (Table 16) and the lower variety of wall lengths (smaller standard deviation in Table 10), which leave fewer feasible combinations and less room for optimization. In Table 19, segment combinations such as Segments 83 and 63 in Case 1 fill a standard panel exactly (combined width 600 mm). Moreover, the height of Body 2 in Case 1 is set to 2470 mm, 30 mm less than the 2500 mm standard-panel height, which is permitted because the height of a PWP panel may be 30 mm shorter than the actual wall height; this height reduction also explains why the usable area of the bodies in Case 2 (Table 18) is 0.7 m2 smaller than the area of the standard panels (Table 17). These observations show that the algorithm exploits the established constraints to achieve targeted optimization.

Table 16

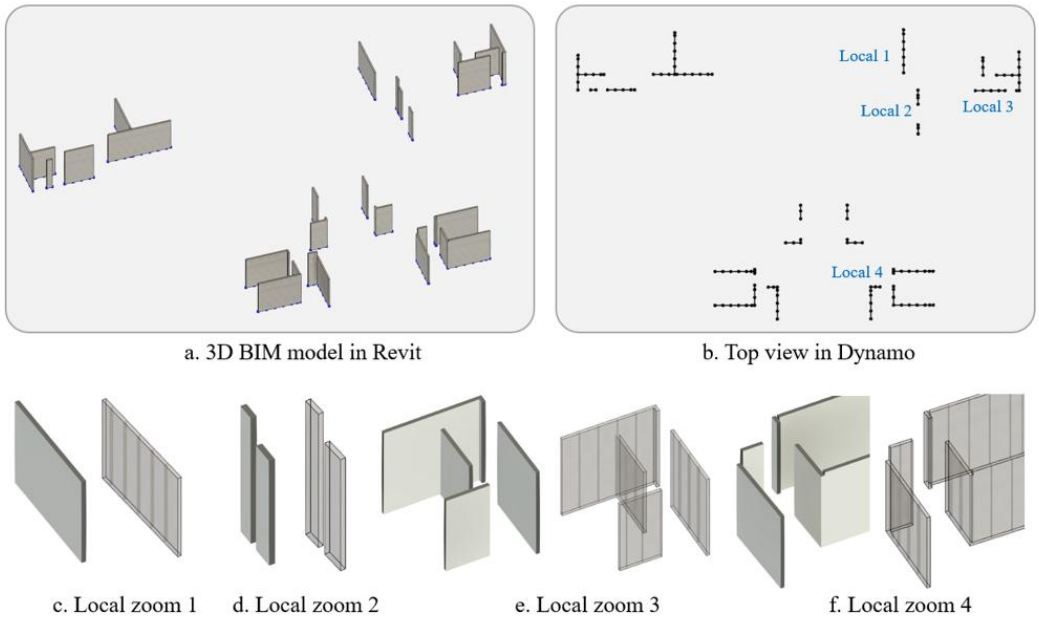
Statistical data of the PWP segments obtained from the first-stage EA

Case	Overall (None, m, m^2)			Width (m)				Height (m)			
	Number	Width	Area	Max.	Min.	Mean	Std.	Max.	Min.	Mean	Std.
1	110	55.55	147.52	0.605	0.200	0.505	0.140	3.00	2.40	2.66	0.21
2	81	39.10	104.91	0.605	0.200	0.480	0.150	3.00	2.50	2.67	0.22

Table 17

Statistical data of the standard PWP panels obtained from the first-stage EA

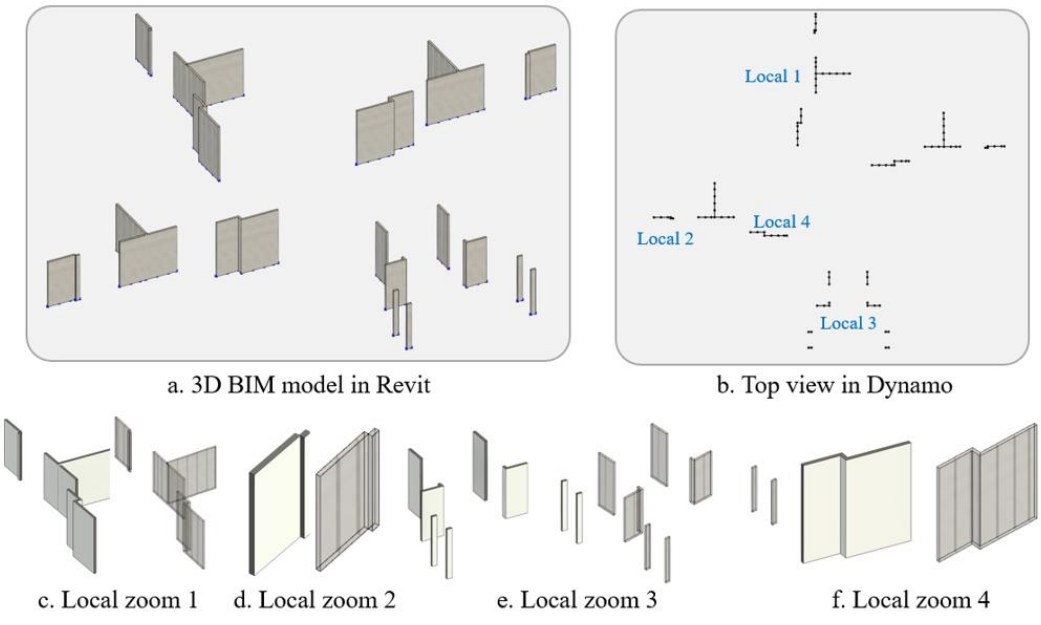
Case	Overall (None, m, m^2, m^2)				Width (m)	Height (m): Number			
	Number	Width	Area	Wasted Area					
1	96	57.6	153.42	5.9	0.6	2.43	2.537	2.630	3.026
2	69	41.4	111.3	6.4	0.6	2.531	2.615	3.023	



a. 3D BIM model in Revit b. Top view in Dynamo

c. Local zoom 1 d. Local zoom 2 e. Local zoom 3 f. Local zoom 4

(a) Case 1

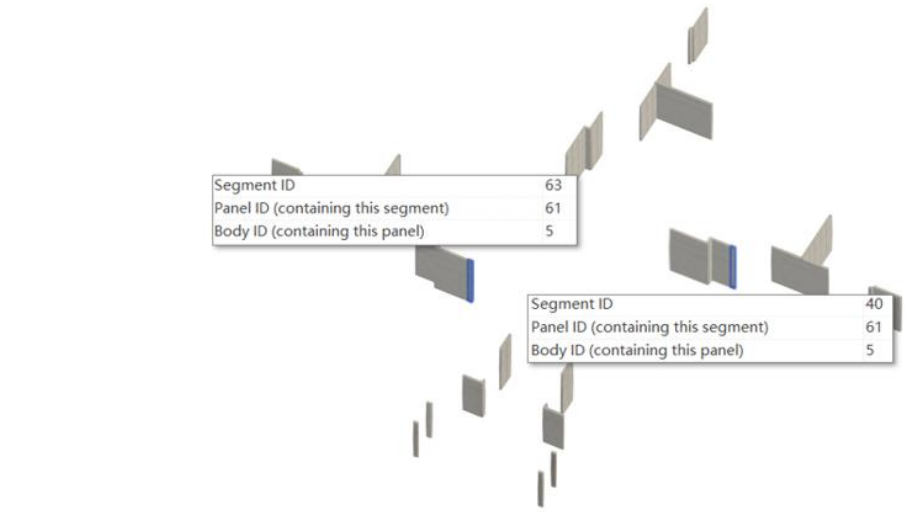


a. 3D BIM model in Revit b. Top view in Dynamo

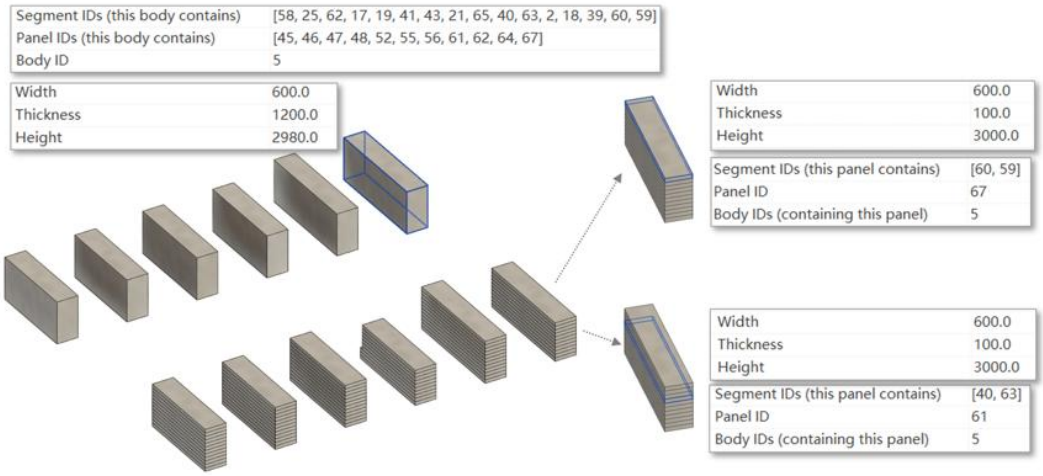
c. Local zoom 1 d. Local zoom 2 e. Local zoom 3 f. Local zoom 4

(b) Case 2

Fig. 21. PWP panel layouts



(a) PWP panel layout



(b) Standard PWP panel combinations and bodies

Fig. 22. BIM models and ID mapping of Case 2

Table 18

Statistical data of the PWP bodies obtained from the second-stage EA

Case	Overall (None, m^2, m^2)		Wasted Area (Nominal area: m^2, Usable area: m^2)		Width (m)	Height (m): Number (Number of remaining standard panels)		
	Number	Nominal area (Usable area)	Vs. Standard panels	Vs. Segments				
1	8	154.5 (154.5)	1.08 (1.08)	6.98 (6.98)	0.6	2.47 3 (0)	2.57 2 (0)	2.97 3 (0)
2	6	115.5 (110.6)	4.2 (-0.7)	10.59 (5.69)	0.6	2.47 2 (0)	2.57 2 (2)	2.98 2 (1)

Table 19

Design results statistics corresponding to Figs. 18 and 22

Case	Body ID (h: mm)	Panel ID	Segment IDs	Segment coordinates ([x, y, x, y, h]: mm)
1	2 (2470)	62	[105]	[10365, 21600, 10970, 21600,2500]
		63	[106]	[10970, 21600, 11575, 21600, 2500]
		64	[107]	[11575, 21600, 12180, 21600, 2500]
		65	[108]	[12180, 21600, 12785, 21600, 2500]
		67	[14]	[18150, 4515, 18150, 4920, 2500]
		68	[83, 63]	[25350, 4515, 25350, 4870, 2500] [29050, 17405, 29050, 17650, 2500]
		69	[44]	[24155, 8550, 24700, 8550, 2500]
		70	[94]	[3910, 21600, 4465, 21600, 2500]
		71	[40]	[19405, 8550, 19950, 8550, 2500]
		73	[109, 15]	[12785, 21600, 13100, 21600, 2500] [18150, 4920, 18150, 5150, 2500]
		74	[79]	[36360, 21500, 36900, 21500, 2400]
		76	[62]	[29050, 16950, 29050, 17405, 2500]
2	5 (2900)	45	[58]	[21860, 18600, 22365, 18600, 3000]
		46	[25, 62]	[3200, 14000, 3400, 14000, 2500] [23205, 18950, 23610, 18950, 3000]
		47	[17]	[14200, 21560, 14200, 22015, 3000]
		48	[19, 41]	[14200, 22300, 14500, 22300, 3000] [11300, 12500, 11300, 12800, 3000]
		52	[43]	[10705, 12800, 11300, 12800, 3000]

55	[21]	[14500, 22905, 14500, 23500, 3000]
56	[65]	[16900, 8905, 16900, 9350, 3000]
61	[40, 63]	[13015, 12500, 13250, 12500, 3000] [23610, 18950, 23850, 18950, 3000]
62	[2, 18]	[15800, 31610, 15800, 31850, 2500] [14200, 22015, 14200, 22300, 3000]
64	[39]	[12510, 12500, 13015, 12500, 3000]
67	[60, 59]	[22600, 18600, 22600, 18950, 3000] [22365, 18600, 22600, 18600, 3000]